MEng and MSc (Engineering) Examination 2025-26

Year 3 / 4
MSc in Advanced Aeronautical Engineering
MSc in Advanced Computational Methods for Aeronautics, Flow Management and Fluid-Structure Interaction

AERO70016 Orbital Mechanics

Coursework assignment

Submission Deadline:
Friday 20th March 2026 at 23:00

The assignment accounts for 15% of the total mark of the module.
The assignment consists of ONE question with TWO parts.
Both a technical report (maximum 5 pages) and the code written for the assignment must be submitted.

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Coursework assignment instructions

Coursework report

Each part of the coursework should be answered in a technical report of 5 pages maximum. Pages exceeding the 5-page limit will not be marked. The report should contain the minimal amount of information required to answer fully each part of the assignment.

- Please include your full name and CID number on the first page of the report, below the title.
- No separate title page, references section, appendices shall be included in the report. The font typeface and size must be legible.
- Word and LaTeX templates for the technical report are provided separately on Blackboard.

Carry-out instructions

The computational tasks in the coursework may be carried out using the NASA General Mission Analysis Tool or MATLAB R2025a with the Aerospace Toolbox. It is also possible to carry out the questions with other toolboxes of your choice in different scientific programming languages (e.g. Python, Julia). Responsibility for the accuracy of the presented results lies with the student.

If you decide to use GMAT for the coursework, please note that its plotting capabilities, while useful for preliminary analysis of the results, are primitive. GMAT-generated figures will not result in full marks. It is possible to significantly enhance the quality of the figures by outputting the results from GMAT into text files and plot the results using your favourite plotting software (MATLAB, matplotlib, gnuplot).

You may find reviewing Figure 5.11 in Section 5.7 of "Orbital Mechanics for Engineering Students", authored by H.D. Curtis, 4th Ed., 2020, useful for answering part (b) of the coursework. The textbook is the core reference of the module and is available through Library Services.

Submission instructions

On Blackboard, please upload in a single zipped archive:

- One PDF file containing the technical report detailing your answers, written using either Microsoft Word or LaTeX.
- A folder containing all code developed to answer the coursework questions.

Coursework question

You are part of the preliminary mission design team for the AURORA mission of the European Space Agency, with the objective of studying the Earth's radiation environment. The mission consists of a spacecraft on a nominal orbit whose orbital elements on 1 January 2026 12:00.000 UTC are

$$a = 26\,600\ \mathrm{km}, \qquad e = 0.7, \qquad i = 116.6°$$
$$\Omega = 115°, \qquad \omega = 90°, \qquad \theta = 180°$$

The AURORA spacecraft has drag and reflectivity coefficients $C_D = 2.2$, $C_R = 1.5$, area $A = 20\ \mathrm{m}^2$, and mass $m = 700\ \mathrm{kg}$.

The spacecraft will be tracked by a ground station at Malargüe, Argentina and a ground station at Kiruna, Sweden. The longitude λ and latitude ϕ of the ground stations in an Earth-Centred, Earth-Fixed (ECEF) frame are for Malargüe:

$$\lambda_1 = 291°, \qquad \phi_1 = -35°,$$

and for Kiruna:

$$\lambda_2 = 20°, \qquad \phi_1 = 67°.$$

Both stations are at sea level.

<u>Part (a)</u>

Propagate the orbit of a spacecraft numerically (i.e., using special perturbations) for a duration of one year.

i. Choose appropriately the numerical method to solve the equations, along with its parameters (e.g. time step, tolerance). [10%]
ii. In addition to the main gravitational acceleration from the Earth's point-mass gravity, define appropriate perturbing accelerations to reproduce the evolution experienced by a real trajectory during this time span, while considering computational efficiency. Your choice may be motivated by examining changes in the history of orbital elements. [15%]
iii. With the choice of physical model in part a.ii, explain the qualitative evolution of a, Ω, ω in an Earth-Centred Inertial frame. Motivate the presence (or absence) of secular, long-periodic, and short-periodic trends. Include plots of the time histories of the orbital elements. [25%]

<u>Part (b)</u>

For each of the ground stations, consider a topocentric reference frame $(O; E, N, Z)$ with its origin at the ground station O, the Z-axis pointing in the same direction as the radius vector from the centre of the Earth to O, the E-axis pointing towards the East direction, and the Z-axis completing the right-handed frame.

The spacecraft is *visible* from a ground station if it is above the horizontal plane of the station. A time interval during which the spacecraft is visible is a *pass*.

i. Propagate the orbit using the physical model chosen in part a.ii. Show, in tabular format, passes of the spacecraft over each ground station between 1 January 2026 12:00:00.000 UTC and 3 January 2026 12:00:00.000 UTC. [10%]
ii. Assess whether passes over the two ground stations are of approximately the same duration. If the duration of the passes over the ground stations differ significantly, explain the different duration through the geometry of the orbit and the location of the ground station. [20%]
iii. Explain whether the conclusions you arrived at in part b.ii would change significantly if the spacecraft were to move according to Keplerian dynamics. [20%]

Assessment criteria

The following criteria will be used to assess whether full marks will be assigned to each of the coursework problems. Please read this carefully.

Coursework report

Full marks for the assignment will be obtained if the report meets the following criteria:

- Demonstrates an excellent understanding of the project assignment. Work presented in the report is necessary and sufficient to answer each of the assignment questions, in other words, all questions are answered fully and no extra work that is irrelevant to the questions asked is presented.
- Methods used and assumptions made are in accordance with the statement of the questions. The rationale for the choice of methods and assumptions is clearly explained in the report.
- All results are presented in the most appropriate form and are accurate. Errors in the calculation of numerical values will be penalised if they result in a significant departure from the correct answer (for example, if a numerical error results in an unphysical answer). All numerical quantities should have appropriate units.
- Demonstrates clear arguments and a well-rounded discussion of the results needed to arrive at each answer. Demonstrates clarity of thought and an excellent defence of the arguments provided.
- The report is correctly formatted. Additional formatting instructions are provided below. Word and LaTeX templates are provided on Blackboard.
- The report is within the page limit (pages exceeding the limit will not be marked).

Formatting instructions

The report title should follow the format “Orbital Mechanics coursework report 2025-26”, and should be placed at the top of the first page. Please write your name and CID number below the title.

The report should be broken into sections and subsections corresponding to each question.

References to equations in text should be numbered (e.g., “see Eq. (11)”). All terms in the equations should be properly formatted.

Any figures should be appropriately numbered and captioned (e.g. “Figure 1: time evolution of the orbital elements.”). References to figures in text should be numbered (e.g. “by visual inspection of Fig. 1”). All plots must be labelled and include units of measure.

No bibliography should be included in the report, unless specified by the coursework instructions.

Computer code

Please include all computer code developed to arrive at the answers in your submission. Source code may form part of assessment in case there is ambiguity in the description of the methods used to arrive at the results, and for plagiarism checks.

Code should be structured to broadly follow the structure of the report, wherever possible. The code should be commented concisely to allow the marker to follow the reasoning employed and clarify the meaning of variables and functions.

Policy on generative AI

Use of generative AI tools is allowed in accordance with the Policy on Generative AI of the Department of Aeronautics.